FULL-STACK (MERN)

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HTML

Definition:- HTML (Hyper Text Markup language) is a text based approach to describing how content contained within a HTML file is structured. (or) It is a standard markup language for creating webpages. HTML describes the structure of webpage

-> This markup tells a webbrowser how to display text, images and otherforms of multimedia on a webpage.

HTML Element:-

An html element is defined by a start tag, somecontent and an end tag.

<tagname > Content ---- </tagname>

The html element is everything from start tag to end tag.

-> Every element consists of an opening tag and a closing tag.

NOTE:- Some HTML elements have no content. (like
 element). These are empty elements. Empty elements do not have a tag.

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Example:

```
<!DocType html>
<html> (<html lang = "en-GB">
<head>
<title> Title of the page </title>
</head>
<body>
<h1> My first heading </h1>
<P> My first paragraph </p>
</body>
</html>
```

(en: English; GB: Greek Britian)

-> <!DOCTYPE> declaration defines that this document is an HTML5 document.

<html> element is the root element of an HTML page.

<head> head element contains meta information about HTML page

<title> element specifies a title for HTML page (which is shown in browser title bar)

<body> element defines the document's body and it is a container for all the visble contents such as headings, paragraphs, images, hyperlinks, tables, lists etc

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<h1> elements define a large heading
<p> element defines a paragraph

HTML Editors:-

1. Write code in Notepad (Windows), TextEdit (Mac)
2. Save the html file with .html extension
3. Set the encoding to "UTF-8".
4. Open the file in browser.

Tip:- We can use .html or .htm as file extension. There is no difference.

The <!DOCTYPE> Declaration:-

-> The <!Doctype> declaration represents the document type and help browser to display web pages correctly.
-> It must only appear once, at the top of the page (before any html tags).
-> The <!DOCTYPE> declaration is note case sensitive.

HTML Headings:-

-> HTML headings are defined with <h1> to <h6> tag.

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→ <h1> defines most important tag.
<h6> defines least important tag.

Ex:

<h1> first heading </h1>
<h2> Second heading </h2>

HTML Paragraphs:-

HTML paragraph are defined with <p> tag.

Ex: <p> This is a paragraph <p>
<p> This is another paragraph <p>

HTML Links:- (Anchor Tag)

HTML links are defined with <a> tag

Ex:

 This is a link.

→ The link's destination is specified in the href attribute.

→ Attribute is used to provide additional information about HTML elements.

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Paragraph Tags

Some of these Tags which may used while writing paragraphs or to highlight text.

1.

strong tag is used to bold the letters

Ex.

 My name is Susmitha

-> previously <b> tag is also used to bold.

2. <u>

To underline an text open and closed </u> tag will be used.

Ex: <u> Susmitha </u>

3.

To italicize any text we use this tag

Ex. <u> Susmitha is a good girl </u>

4. Subscript:-

To write Subscript <sub> tag is used

Ex. log ₁₀

O/p : $\log_{10}$

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5. **Superscript**:-

To give Superscript <sup> tag will be used

Ex: x ²

o/p - x^2

6. **Strike through**:-

To strike any particular word we use this tag.

Ex: I am <del> good </del> bad girl .

Note: List of tags will be available in w3 schools.

Boilerplate of HTML:- (Template / structure)

```
<!DOCTYPE html>
<html lang = "en-GB">
 <head>
    <meta charset = "UTF-8"> --> Meta is an
 <title> Title </title>          hidden information
 </head>
 <body>
    <h1> My heading </h1>
 </body>
 </html>
```

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Meta Tag:-

Meta information is nothing but hidden info.. Meta tag always be inside the head element and used to specify character set, page description, keyword, etc...

-> charset UTF-8 it is a meta tag which indicates characters correctly in web page. Example if we use emoji, #code etc.. it is displayed correctly using ascii on browser.

UTF - UNICODE TRANSFORMATION FORMAT

HTML void elements:-

HTML elements that don't have closing tags such as
, <hr/> and <img> tags are considered as "empty tags", void tags or self closing tags as they cannot contain any content

Ex:

```
<p> Twinkle Twinkle <br/> (</br> tag breaks the line)
    little star <br/>
    how I wonder <br/>
    what you are <br/> </p>
```

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<hr/>

It is a void tag which creates a line.

Ex: <h1> Cells Basic Unit </h1>
<hr/>

Lists in HTML:-

HTML allows web developers to group a set of related items in lists.

It is of 2 types:-

1. **Ordered Lists**

An ordered list starts with "<ol>" tag. Each list item starts with "<li>" tag

-> The list items will be marked with numbers by default.

Ex:
```
<h1> Menu </h1>
<ol>
  <li> Coffee </li>
  <li> Tea </li>
  <li> Milk </li>
  <li> Boost </li>
</ol>
```

O/p:

1. Coffee
2. Tea
3. Milk
4. Boost

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2. Unordered list:-

An unordered list starts with <ul> tag. Each list item starts with <li> tag.

-> The list items will be marked with bullets (small black circles) by default.

Ex:

```
<ul>
    <li> Coffee </li>
    <li> Tea </li>
    <li> Milk </li>
</ul>
```

o/p:
- Coffee
- Milk
- Tea

3. Nested lists

A nested list is a list inside another list. You can create a nested unordered list, or a nested unordered list or even an ordered list nested inside an Unordered one.

Ex:

```
<ol>
    <li> Coffee </li>
    <li> Tea </li>
        <ul>
            <li> Green Tea </li>
            <li> Black Tea </li>
        </ul>
```

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```
<li>Milk</li>
</ol>
```

o/p:-

1. Coffee
2. Tea
   - Green Tea
   - Black Tea
3. Milk

Changing as Roman Numbers in Ordered list.

In ordered list if we have to use A, B, C, D or i, ii --- instead of 1, 2, 3, 4 we use the following method by inserting type in <ol> tag.

Ex

```
<ol type = "a">
    <li>Tea</li>
    <li>Milk</li>
</ol>
```

o/p: a. Tea
b. Milk

-> Similarly you can use, i, ii -- or a, b -- or A, B -- or I, II -- by putting the first letter in type under ol tag

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→ To change in nested list type.

Ex:
```
<ol type="I">
    <li> Tea </li>
        <ol type="i">
            <li> Green Tea </li>
            <li> Black Tea </li>
        </ol>
    <li> Coffee </li>
    <li> Milk </li>
</ol>
```

o/p:

I. Tea
    i. Green tea
    ii. Black tea
II. Coffee
III. Milk

4. HTML Description lists:-

HTML also supports description list

A description list is a list of terms, with a description of each term.

→ The <dl> tag defines the description list, the <dt> tag defines the term (name) and <dd> tag describes each term.

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Ex:

```
<dl>
    <dt> Coffee </dt>
    <dd> -black hot drink </dd>
    <dt> Milk </dt>
    <dd> - white cold drink </dd>
</dl>
```

o/p: Coffee

- Black hot drink

Milk

- white cold drink

HTML List Tags:-

	- Unordered list
	- Ordered list
	- list item
<dl>	- description list
<dt>	- Defines a Term in description list
<dd>	- Describes the term in description list.

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HTML Anchor Tag:-

HTML links are defined with <a> tag

Ex: Anchor Tag open — Anchor Tag close

Visit Prepinsta prime!.

Destination URL — Hyperlinked Content

-> href = hyperlink reference, it is an attribute (Destination URL is the value for the attribute).

Ex:

I use Prepinsta to prepare for placement

o/p:

I use Prepinsta to prepare for placements

This word contains URL of the page.

Attribute:-

href is an attribute; Similarly an html element contains multiple attributes.

-> <tag attribute = "value 1" attribute 2 "value 2" > Content </tag>.

Ex: I use prepinsta to prepare placements.

o/p: I use Prepinsta to prepare placement

Target = _blank is used to open URL in new tab by default

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We can use anchor tag in lists also

Ex: <h2> My Social profile <h2>

  Linkeding

  Instagram

Image Tag

Image tag is used to insert images.

It is an self closing tag or Void tag.

↓ Selfclosing | image tag. | ↓ image URL

Attribute

Tip: Create a new folder in vs code with images

1. Copy the path paste it in src

Ex:

To align image in centre.

To align height & width :-

(auto)

auto → takes auto width

alt :-

It is an alternative attribute which appear if in case image is failed to load.

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Ex:

<img src = "http image. com" all = "My Image" >

o/p:

If image failed to load then text "My Image" will appear.

HTML Attributes

HTML attributes provide additional information about HTML elements.

→ All HTML elements can have attributes.

→ Attributes provide additional information about elements.

→ Attributes are always specified in start tag.

→ Attributes usually come in name/value pairs like name = "value".

Ex: Prepinsta

href → name

"www.prep.com" → value.

1. href attribute:-

The <a> tag defines a link/hyperlink. The href attribute specifies the URL of the page link goes to:

Ex: Instagram

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2. Src Attribute

The <img> tag is used to embed an image in an HTML page. The "src" attribute Specifies the path to the image displayed.

Ex:

<img src = "img_girl.jpg>

→ There are two ways to specify the URL in the src attribute.

1. Absolute URL:

Links to an external image that is hosted on another website.

Ex: <img srce = "https://www.prepinsta.com" >

- Notes:

External images might be under 'copyright. If you do not get permission to use it, you may be in Violation of copyright laws.

2. Relative URL:-

links to an image that is hosted within a website. Here URL doesnot include domain name. If the URL begins without a slash, it will be relative to current page.

Ex: 'src: " img_girl.jpg"

If the URL begin with a slash, it will be

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relative to domain Ex: src = "/images/img-girl.jpg".
-> For this we are going to create a file in folder with images.

Note:-

3. Width and height attributes:-

The <img> tag contains the width and height attributes, which specifies the width and height of the image in pixels.

Ex: <img src = "img.jpg" width = "500" heigh = "200px" >

4. alt Attribute:-

The alt attribute for <img> tag specifies an alternate text for an image, if the image for some reason cannot be displayed. This can be due to a slow connection or some reason in src attribute.

Ex:

<img src = "img-girl.jpg" alt = "Girl with coat" >

o/p :- Girl with coat (if the image src fail to load)

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5. Style attribute:-

The style attribute is used to add styles to an element, such as color, font, size and more.

Ex:

```
<p style = "color:red;">This is red paragraph</p>
```

6. The lang attribute:-

You should always include the lang attribute inside the <html> tag, to declare the language of web page. This is meant to assist search engines and browsers.

Ex:

```
<!DOCTYPE html>
<html lang="en">
<body>
  ...
</body>
</html>
```

→ Country code can also be added to the language code in lang attribute.

Ex: `<html lang = "en-US">`

→ first two characters define the languag & last two characters define country.

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HTML Styles:-

The HTML style attribute is used to add styles to an element, such as color, font, size and more.

HTML Style attribute:-

Setting the style of an HTML element, can be done with style attribute.

The HTML style attribute has following syntax.

** <tagname style = "property: value;">

→ The property is css property

→ The value is css value.

Background colour:-

To set background color for a page

The css background-color property defines the background color for an HTML element.

Ex:

```
<body style="background-color:powderblue;">
<h1>heading</h1>
</body>
```

(background-color: css property; powderblue: css value)

To set Background color for two different elements:-

```
<body>
<h1 style="background-color:powderblue;">
This is a heading</h1>
```

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<p style background-color: tomato; "> This is a paragraph

</body>

Text Color:-

The css color property defines the text color.

<h1 style = "color: blue; "> This is heading </h1>

 Paragraph

Fonts

The css font family property defines the font to be used for an html element.

Ex:

<h1 style = "font-family: verdana; "> Heading 1 </h1>

<p style = "font- " : courier; "> Paragraph 1

Text Size:

The css font size property defines the text size for an HTML element.

Ex:

<h1 style = "font-size: 300%; "> Heading 1 </h1>

 Paragraph 1

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Text Alignment:

The css text-align property defines the horizontal text allignment for an HTML element.

Ex:

```
<h1 Style = "text-align:center;"> Centered Heading </h1>
<p Style = "text-align:center;"> Centered Paragraph </p>
```

HTML Tables

HTML tables allows web developers to arrange data into rows and columns

→ A table in HTML consists of table cells inside Rows and columns.

Ex:

```
<body>
<table>          Not necessary to have this tag
  <thead> →
  <tr> → Row
    <th> Company </th> → <th> stands for heading
    <th> Contact </th>
    <th> Country </th>
  </tr></thead>
    <tbody>
  <tr>
    <td> Google </td>
    <td> Mary </td>
```

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```
    <td> USA </td>
  </tr>
  <tr>
    <td> Microsoft </td>
    <td> Sony </td>
    <td> India </td>
  </tr>
</tbody>
</table>
```

td -> Table description

Table cells:-

Each table cell is defined by <td> and </td> tag

-> td stands for table data

-> Everything between <td> and </td> are content of table.

NOTE:- A table cell contain all sorts of HTML elements: text, images, lists, links etc.

Header:-

<th> stands for table header.

By default header is bolded & centered, but we can change it with CSS.

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-> For above example o/p will be without table border to insert border

Ex: <table border = "1">

-> To make table width to full screen

<table border = "1" width = "100%">

Table Rowspan & Colspan:-

Rowspan:-

To make a cell span over multiple rows, use the rowspan attribute.

Ex:

```
<table border = "1" width = 100%>
    <tr>
        <th> Month </th>
        <th> Reason </th>
        <th> Amount </th>
    </tr>
    <tr>
        <td rowspan = "3"> Jan </td>
        <td> Party </td>
        <td> ₹5000 </td>
    </tr>
    <tr>
        <td> Study </td>
```

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```
        <td> 4000 </td>
    </tr>
    <tr>
        <td> Study </td>
        <td> 6000 </td>
    </tr>
    <tr>
        <td> Feb </td>
        <td> Party </td>
        <td> 5000 </td>
    </tr>
</table>
```

Colspan:-

```
<table  border="2" >
<tr>
    <th colspan="2"> Name </th>
    <th> Age </th>
</tr>
<tr>
    <td> Kaushik </td>
    <td> Singh </td>
    <td> 26 </td>
</tr>
</table>
```

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HTML Title & FavIcon:-

A favicon is a small image displayed next to the page title in browser tab.

-> We can use any image as favicon. It is displayed in browsers title bar. or

-> To add fav icon

add a <link> element after to <title> element

Ex:

```
<title>My page</title>
<link rel="icon" type="image/x-icon"
                 href="/image.jpg">
</head>
```

Tip:- Different icons can be downloaded from website flaticon.com or browser

HTML Forms:-

A HTML form is used to collect user input. The user input is most often sent to server for processing.

-> The <form> element is used to create an HTML form for user input.

-> <input> element is most used form element along with type attribute.

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HTML input Types:-

There are different types of input types we can use in HTML.

```
<input type="button">
<input type="checkbox">
<input type="textfield">
<input type="text">
<input type="Radio">
```

Ex:

```
<form action="#">
    <input type="text" placeholder="Enter your
                                    firstname"></br>
<input type="text" placeholder="Enter your
                                    last name"><br/>
<input type="text" Placeholder="Enter your
                                    Mail id">
<textarea name="address" id="address"
        cols="30" rows="10" placeholder="Enter your
                    Address"></textarea>
Gender </br>
<input type="radio" value="Male" names="gender">Male
<input type="radio" value="Female" names="gender">Female
</br>
<input type="checkbox" name="car" id="car">Car
```

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```
<input type="checkbox" name="bike" id="Bike">Bike
<input type="Submit" value="Submit">
</form>
```

Note:- Default width of an input field is 20 characters.

The <label> Element:-

The <label> tag defines a label for many elements.

-> The <label> element is useful for screen-reader uses, because the screen-reader will read out the label when the user focuses on the input element.

Ex:

```
<form>
<label for="fname">First Name:</label><br>
<input type="text" id="fname" name="fname"><br>
<label for="lname">last name:</label><br>
<input type="text" id="lname" name="lname">
</form>
```

Tip:- Different types of input types will be available in w3schools (syntax).

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HTML Form Get & Post:-

The form data can be sent as URL variables (method = "get") or as an HTTP post transaction (method = "post")

Get Method

This method is useful for form submission where a user want to bookmarks the result.

-> There is a limit how much data you can place in a URL (varies between browsers) therefore you cannot be sure that all of the form data will be correctly transferred.

-> Never use the "get" method to pass sensitive information (passwords) bcoz it will be visible if we click on browser URL.

Ex.

<form action = "a" method = "get">

-> If we give get method & submit the answers. then if we click on URL the answers will be visible.

Post Method:-

This method sends the form data as an HTTP transaction.

-> Form submissions with the post method cannot be bookmarked or visible.

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-> Post method is more secure than Get
-> Post doesnot have size limitations like Get

Ex:

```
<form action:"#" method:"post">
```

EMail using HTML forms:-

We can send the mail directly using HTML forms. The syntax is

```
<form action="mailto:susmithr1810@gmail.com"
      method:"post" enctype="text/plain">
```

-> Write the remaining code of form input & Submit and run it on browser (after click on submit it directly leads to mail page).

Enctype attribute:-

The input formenctype attribute specifies how the form data should be encoded (to mail) when Submitted (only for forms with method="post").

To add subject in mail:-

put ? mark after mail as shown in below Syntax.

Ex:

```
<form action="mailto:abc@gmail.com?Subject=Student form
      Data" metho="post" enctype "text/plain">
```

o/p:-

Subject will be printed

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To include Body in Mail:-

To print body of the mail use the following Syntax:-

<form action="mailto:abc@gmail.com?Subject=Write Subject of mail her &Body=Write body here" method="post" enctype="text/plain" >

-> & This Symbol is used to include body after subject.

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